

LogLens

Transforming Raw Web-Server Access Logs into Actionable Security Insights

An automated threat intelligence and log monitoring platform powered by Python, Flask, React, and signature-based regex engines paired with behavioral detection algorithms.

PRESENTED BY

Gaurav Jagadhane

INTERNSHIP ORGANIZATION

Persevex

ACADEMIC YEAR

2026-27

PROJECT TYPE

Minor Project Report

⚠ Problem Statement



The Challenge with Web Logs

Web servers (Apache/Nginx) generate continuous streams of high-volume access-log entries. Manual inspection of raw log files is extremely slow, tedious, and error-prone.

Malicious actors often hide attack patterns (such as stealthy payload injection or distributed probes) within thousands of legitimate user requests, making timely threat detection impossible without automation.



Target Attack Vectors

LogLens specifically addresses critical OWASP Top 10 web vulnerabilities:

- ✓ **SQL Injection (SQLi):** Malicious database query manipulation
- ✓ **Cross-Site Scripting (XSS):** Client-side script injection
- ✓ **Directory Traversal:** Unauthorized path manipulation (`../`)
- ✓ **Brute Force Attacks:** Repeated HTTP 401 Unauthorized attempts

Proposed Solution — LogLens Workflow

LogLens provides an end-to-end automated pipeline that ingests unformatted server logs, isolates threat vectors, presents interactive visualizations, and produces actionable security reports.



🎯 Project Objectives



1. Log Processing Objectives

- ✓ **Automated Ingestion:** Parse standard Apache and Nginx web server access logs seamlessly.
- ✓ **Field Extraction:** Extract Client IP, Timestamp, HTTP Method, Requested URL/Path, Protocol, Status Code, and Payload Size.
- ✓ **Progressive Processing:** Process records line-by-line to maintain high performance with larger files.
- ✓ **Fault Tolerance:** Safely bypass malformed log entries without interrupting overall analysis.



2. Security Analysis Objectives

- ✓ **Multi-Threat Detection:** Identify SQLi, XSS, Directory Traversal, and Brute Force patterns accurately.
- ✓ **Behavioral Correlation:** Monitor repeated authentication failures (HTTP 401) from single source IPs.
- ✓ **Severity Tagging:** Categorize threat events with standard risk levels (HIGH, MEDIUM, LOW).
- ✓ **Report Generation:** Provide exportable CSV spreadsheets and formatted PDF summaries.

System Architecture



Frontend Layer

React + Vite + Recharts

- ✓ Log File Upload Drag & Drop Interface
- ✓ Real-time Metrics Dashboard
- ✓ Recharts Visual Threat Graphs
- ✓ Interactive Event Filtering Table
- ✓ Trigger CSV / PDF Downloads



Backend API Layer

Python + Flask REST API

- ✓ File Ingestion & Stream Controller
- ✓ Regex Parsing Engine Integration
- ✓ Threat Detection & Aggregator
- ✓ Report Generation Service
- ✓ REST Endpoint Routes (/analyze, /export)



Analysis & Output Engine

Regex + ReportLab

- ✓ Signature Patterns (SQLi, XSS, Path Traversal)
- ✓ Stateful 401 Behavioral Counter
- ✓ Severity Classifier Algorithm
- ✓ ReportLab PDF Canvas Renderer
- ✓ Structured CSV Data Formatter

</> Log Parsing Module

The parser converts unstructured raw log strings into structured, queryable JSON objects using optimized Regular Expressions.

RAW ACCESS LOG RECORD EXAMPLE:

```
192.168.1.10 - - [13/Jul/2026:09:15:21 +0530] "GET /login HTTP/1.1" 200 512
```

EXTRACTED STRUCTURED SCHEMA:

CLIENT IP

192.168.1.10

TIMESTAMP

13/Jul/2026:09:15:21

HTTP METHOD

GET

REQUEST URI / PATH

/login

HTTP STATUS

200 OK

PAYLOAD SIZE

512 Bytes

Threat Detection Engine

Attack Category	Detection Mechanism / Regex Signature	Severity	Risk Description
SQL Injection (SQLi)	' OR 1=1 , UNION SELECT , DROP TABLE	HIGH	Unauthorized access, data exfiltration or database destruction
Cross-Site Scripting	</div> , javascript: , onerror=	HIGH	Session hijacking and malicious client-side script execution
Directory Traversal	.. / , .. %2f , /etc/passwd	HIGH	Arbitrary file reading outside web root directory
Brute Force Attack	Threshold counter for consecutive 401 Unauthorized responses per IP	HIGH	Automated password guessing / credential stuffing attacks

Threat Tagging & Event Lifecycle

Detection Processing Pipeline

1. Parse Raw HTTP Request String
2. Evaluate Regex Signatures & Stateful Counters
3. Match Triggered → Assign Severity Tag & Threat Type
4. Construct Normalized Security Threat Event Object

Structured Security Event Output

```
{ "type": "sql_injection", "severity": "HIGH", "client_ip":  
"198.51.100.23", "timestamp": "2026-07-26T14:22:10Z", "method":  
"GET", "path": "/product?  
id=1%20UNION%20SELECT%20pass%20FROM%20users", "http_status":  
403, "action_taken": "Logged & Alerted" }
```


Behavioral Detection — Brute Force



Observed Log Sequence

```
203.0.113.64 - - "POST /login HTTP/1.1" 401 128 203.0.113.64 - -  
"POST /login HTTP/1.1" 401 128 203.0.113.64 - - "POST /login  
HTTP/1.1" 401 128 203.0.113.64 - - "POST /login HTTP/1.1" 401 128  
203.0.113.64 - - "POST /login HTTP/1.1" 401 128
```

Single signatures cannot detect this attack. Multi-record stateful evaluation is mandatory.



Detection Logic Algorithm

- ✓ **State Grouping:** Filter and group failed requests by Client IP address.
- ✓ **Response Filtering:** Isolate requests returning `401 Unauthorized` responses.
- ✓ **Time Windowing:** Count failed login attempts occurring within a configurable time window.
- ✓ **Threshold Trigger:** Flag source IP as **Brute Force Attacker** once threshold is exceeded.

Interactive Security Dashboard

SIEM-LITE SECURITY MONITORING

LogLens

Turn web-server access logs into practical security insights.

Upload Log

Load Demo

Analysis completed successfully.

Total Requests

10

Total Threats

4

Unique Attackers

4

Malformed Lines

0

Source: **demo_access.log**

Export CSV

Export PDF

Threat Investigation & Analysis

The LogLens dashboard enables SOC analysts to move smoothly from macro security indicators to granular event forensics.

Threat Investigation Table Fields

- **Threat Type:** Classified category
- **Severity Level:** Risk rating
- **Source IP:** Attacker origin
- **Timestamp:** Exact UTC time
- **HTTP Status:** Response code
- **Target URI:** Malicious payload

Attack Timeline Tracking

Correlates request timestamps to reveal spike patterns and automated scanner activity across time intervals.

Multi-Vector Filtering

Allows instant searching and filtering by specific attack types, IP ranges, or HTTP status codes.

Report Generation Module



CSV Spreadsheet Export

- ✓ **Structured Row Data:** Tabular records containing all parsed log attributes and threat flags.
- ✓ **SIEM & Excel Compatibility:** Easy ingestion into external SIEM platforms, Excel, or Python Pandas.
- ✓ **Granular Forensic Storage:** Ideal for deep offline data manipulation and audit logging.



PDF Executive Report

- ✓ **ReportLab Engine:** Dynamically generated PDF documents with clean layout formatting.
- ✓ **Executive Summary:** High-level metrics charts and threat distribution breakdowns.
- ✓ **Submission Artifact:** Professional artifact suitable for management reviews and academic evaluation.

Technology Stack

FRONTEND DEVELOPMENT

React + Vite

Fast SPA UI rendering, modular component hierarchy, and responsive styling.

DATA VISUALIZATION

Recharts Library

Interactive threat distribution charts, time-series graphs, and status metrics.

BACKEND FRAMEWORK

Python + Flask

Lightweight RESTful API framework handling file ingestion and analytical routing.

PATTERN DETECTION

Regular Expressions

High-performance string matching rules for identifying signature threat payloads.

PDF GENERATION

ReportLab Library

Programmatic document rendering engine producing formatted PDF summaries.

TESTING & VERSIONING

pytest & Git

Automated unit test suite validating parser accuracy and version management.

Testing & Validation Results

Rigorous automated unit testing was conducted using `pytest` against synthetic and real-world web server access logs.

Test Scenario	Input Condition	Expected Result	Status
Normal Traffic Verification	Standard HTTP GET / POST logs	No false positive threat alerts	✓ PASSED
SQL Injection Suite	URL parameters containing <code>UNION SELECT</code>	Accurately flagged as SQLi (HIGH)	✓ PASSED
Directory Traversal Suite	Payloads with <code>../ ../etc/passwd</code>	Accurately flagged as Traversal	✓ PASSED
Brute Force Behavior	5+ consecutive HTTP 401s from single IP	Brute Force event generated	✓ PASSED
Malformed Record Safety	Corrupted lines & incomplete headers	Skipped gracefully without application crash	✓ PASSED

 **FINAL VERIFIED SUMMARY:**

15 / 15 Unit Tests Passed • 1,250 Log Entries Processed • 48 Threats Identified • 0 Application Crashes

💡 Challenges & Key Learning Outcomes

Engineering Challenges

- ✓ **Log Diversity:** Handling subtle format differences between Nginx and Apache Combined access logs.
- ✓ **Unstructured Data:** Crafting resilient regex patterns that resist malformed inputs.
- ✓ **Full-Stack Integration:** Managing asynchronous file upload states between React and Flask backend endpoints.
- ✓ **Detection Precision:** Minimizing false positives while maintaining broad threat detection coverage.

Key Learning Outcomes

- ✓ **Advanced Regex & Parsing:** Gained hands-on experience in string manipulation and log restructuring in Python.
- ✓ **RESTful API Architecture:** Mastered Flask request lifecycle management and data serialization.
- ✓ **Full-Stack Development:** Deepened proficiency in React state management and Recharts data visualization.
- ✓ **Security Fundamentals:** Strengthened understanding of OWASP threat vectors and log forensics.

Limitations & Future Scope



Current Scope Limitations

- ✓ **Log Formats:** Currently optimized for standard Apache and Nginx web access logs.
- ✓ **Static Parsing:** Relies on signature-based rules and basic stateful counters rather than machine learning.
- ✓ **Passive Monitoring:** Provides detection and reporting without automated firewall response or IP blocking.



Future Enhancements

- ✓ **Real-Time Streaming:** Ingest live log streams using WebSockets or syslog forwarders.
- ✓ **Geo-IP Enrichment:** Map attacker IP addresses to geographic locations on an interactive globe.
- ✓ **Machine Learning Detection:** Integrate unsupervised anomaly detection for zero-day threat discovery.
- ✓ **Threat Intelligence Feeds:** Cross-reference IPs with global blacklists (e.g., AbuseIPDB).

Conclusion & Summary



Automating Security Log Analysis

LogLens successfully demonstrates a practical, SIEM-lite web application that bridges raw log parsing, signature & behavioral threat detection, interactive visual analytics, and exportable security reports into a single unified platform.

✓ Automated Ingestion

✓ Real-Time Detection

✓ Exportable Reports



Thank You!

Questions & Discussion

I sincerely appreciate the opportunity to present the LogLens project. I welcome any questions, suggestions, or feedback.

Presenter: Gaurav Jagadhane

Internship: Persevox

Project: LogLens SIEM-Lite